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Pyroxene mineral chemistry of the Jinchuan intrusion, China

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With 6 Figures

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Summary

Chemical compositions of orthopyroxene and clinopyroxene from the Jinchuan ultramafic intrusion have been obtained by electron microprobe analysis. The Mg′ number (MgO/(MgO + FeO)) for both pyroxenes falls within narrow ranges, 82–87 for clinopyroxene and 81–85.5 for orthopyroxene, suggesting limited magma differentiation in regard to the present igneous body. The Al_2O_3 content ranges from 2.44 wt.% to 4.43 wt.% and increases with decreasing Mg′ of the pyroxenes, i.e., with the more evolved magma. This is attributed to the relatively greater effects of Al_2O_3, TiO_2, Cr_2O_3 and Fe_2O_3 than that of SiO_2 on pyroxene crystallization.

Negative linear relationships between Ti^{4+} and Si^{4+}, and Al^{3+} and Si^{4+} characterize the pyroxenes. In clinopyroxene, regression of Si^{4+} versus Al^{3+} results in a straight line with a slope of -1.012, indicating that the decrease of Si^{4+} in the crystal structure is matched by an increase only in tetrahedral Al^{3+}; octahedral Al^{3+} has remained relatively constant. The negative linear relationship between Ti^{4+} and Si^{4+} in clinopyroxene reflects either a greater tendency of Ti^{4+} to occupy octahedral sites than Al^{3+}, or that replacement of Al^{3+} for Si^{4+} demands a more efficient charge balance. The scatter in plots of Ti^{4+} versus Si^{4+} for orthopyroxene indicates that charge balance is not as critical as structure symmetry.

The crystallization temperature of pyroxene is calculated to be 1108–1229 °C using *Wood* and *Banno*'s (1973) two pyroxene thermometer, and is within 40 °C of that calculated from *Wells*'s (1977) thermometer. The distribution coefficient (Kd) for Mg^{2+} and Fe^{2+} between clinopyroxene and orthopyroxene is estimated to be 0.86, which is higher than that of the other intrusions and lower than that of mantle nodules, but still falls within their Kd-1/T trend. This suggests that the Kd value of pyroxene is controlled mainly by temperature.

Zusammenfassung

Mineralchemie der Pyroxene der Jinchuan-Intrusion, China

Die chemische Zusammensetzung von Orthopyroxenen und Klinopyroxenen aus der ultramafischen Jinchuan Intrusion wurden mit der Mikrosonde bestimmt. Die Mg-Zahl (MgO/(MgO + FeO)) beider Pyroxene liegt innerhalb enger Grenzen, 82–87 für Klinopyroxen und 81–85.5 für Orthopyroxen. Dies weist auf beschränkte magmatische Differentiation der Intrusion hin. Der Al_2O_3-Gehalt liegt zwischen 2.44 Gew.% und 4.43 Gew.% und nimmt mit der abnehmenden Mg-Zahl der Pyroxene ab, d.h. mit dem mehr entwickelten Magma. Dies wird damit erklärt, daß Al_2O_3, TiO_2, Cr_2O_3 und Fe_2O_3 einen größeren Einfluß auf die Kristallisation der Pyroxene ausüben als SiO_2.

Die Pyroxene werden durch negative lineare Beziehungen zwischen Ti^{4+} und Si^{4+}, sowie Al^{3+} und Si^{4+} charakterisiert. In Klinopyroxenen resultiert die Regression von Si^{4+} gegen Al^{3+} in einer geraden Linie mit einer Neigung von -1.012. Dies weist darauf hin, daß die Abnahme der Si^{4+} Gehalte in die Kristallstruktur durch Zunahme von ausschliesslich tetraedrischem Al^{3+} kompensiert wird; oktaedrisches Al^{3+} ist relativ konstant geblieben. Die negative lineare Beziehung zwischen Ti^{4+} und Si^{4+} in Klinopyroxenen geht entweder auf eine stärkere Tendenz des Ti_4O_2, oktaedrische Plätze zu besetzen zurück, oder darauf daß ein Ersatz von Al^{3+} für Si^{4+} einen effizienteren Ladungsausgleich verlangt. Die unregelmäßige Verteilung der Plots von Ti^{4+} gegen Si^{4+} in Orthopyroxenen läßt erkennen, daß Ladungsausgleich hier nicht so kritisch ist wie die Symmetrie der Struktur.

Die Kristallisationstemperatur der Pyroxene wurde mit dem Zwei Pyroxenthermometer nach *Wood* und *Banno* (1973) mit 1108–1229 °C bestimmt. Diese Werte liegen innerhalb von 40 °C des von *Wells* (1977) berechneten. Der Verteilungskoeffizient (Kd) für Mg^{2+} und Fe^{2+} zwischen Klinopyroxen und Orthopyroxen wird auf 0.86 berechnet; das ist höher als der aus anderen Intrusionen und niedriger als der von Mantelxenolithen, fällt aber immer noch innerhalb des Kd-1/T Trends derselben. Dies legt den Gedanken nahe, daß der Kd Wert der Pyroxene hauptsächlich durch Temperatur bestimmt wird.

Introduction

The Jinchuan intrusion is a dyke-like igneous body that hosts a large Ni–Cu sulfide deposit (*S.G.U.*[1], 1984; *Chai* and *Naldrett*, 1992a). The intrusion is divided into three subchambers, the west, west-central and east, each of which hosts a major ore body (*Chai* and *Naldrett*, 1992b). In the west and west-central subchambers, a lateral zonation of rock types is present, with dunite at the core, grading through lherzolite to olivine pyroxenite at the margins. The eastern subchamber is characterized by vertical layering in which dunite occurs at the base and grades upwards into lherzolite, plagioclase lherzolite, and then lherzolite and olivine pyroxenite at the top.

Chai and *Naldrett* (1992b) observed negative linear relationships between MgO and most other components with the exception of a few mobile components such as Na_2O and K_2O, and trace elements such as Sr and Rb and indicated that the rocks were essentially a mixture of cumulus olivine and the primary magma. The cumulus olivine has forsterite content mainly between 83–85%, which corresponds to a calculated MgO/(MgO + FeO) of the primary magma of 0.64. This ratio, combined with the FeO content of the rocks, gives a MgO content of about 12 wt.% for the

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Jinchuan magma. The present igneous body represents only the ultramafic cumulates of the primary Jinchuan intrusion (*Chai* and *Naldrett*, 1992b).

Both orthopyroxene and clinopyroxene are ubiquitous in the Jinchuan intrusion, but they generally constitute less than 40 vol.% of the rock. The abundance of these minerals increases from the core to the margins of the intrusion, corresponding to variation of rock types from sulfide-bearing dunite, to lherzolite and olivine pyroxenite. The pyroxenes occur almost everywhere as interstitial minerals to cumulus olivine, and in places as reaction rims on olivine. The reaction rims consist most commonly of orthopyroxene, or of an inner ring of orthopyroxene and an outer ring of clinopyroxene. In sulfide-bearing dunite, small amounts of pyroxene occur as discrete aggregates or irregular crystals with sulfides in the interstitial voids, and in many places are cut by sulfide veinlets. In the less olivine-rich rocks, pyroxenes up to 8 mm in size exist as oikocrysts enclosing small to medium sized euhedral olivine crystals. No exsolution textures of either clinopyroxene in orthopyroxene, or vice versa, as commonly observed in pyroxenes from other intrusions (*Atkins*, 1969; *Barnes*, 1983; *Wilson*, 1982), appear to be present in the Jinchuan rocks. Pyroxenes are commonly altered in the more mafic or strongly sulfide-mineralized rocks. The alteration minerals in most cases are tremolite, serpentine and chlorite. Lizardite is the most common serpentine mineral in the altered rocks; it typically replaces olivine and orthopyroxene as bent tabular aggregates. Chlorite occurs mainly as tiny elongated crystal aggregates or as radial clusters that have replaced interstitial pyroxene.

In the present study, fresh pyroxene grains, in particular from samples with both clinopyroxene and orthopyroxene, were selected from polished thin sections. Twenty three grains of clinopyroxene in 12 sections and 12 grains of orthopyroxene in 9 sections were analyzed using the electron microprobe. The purpose was to characterize the chemical features of pyroxenes, to analyze various factors that control the abundance of the major and trace elements, and to calculate their crystallization temperatures. The close association of the pyroxenes with sulfide mineralization was used to examine the effect of sulfur and oxygen fugacities on the distribution coefficient of Mg^{2+} and Fe^{2+} between clino- and orthopyroxene.

Chemical composition of pyroxenes

Analyses of pyroxenes were conducted on the Cameca-SX50 electron microprobe in the Department of Geology, University of Toronto. A 15 kv accelerating voltage, a 30 nA beam current and a scanned electron beam (5 μm) were used. Synthetic TiAl pyroxene glass was used as a standard for Ti and Al; natural wollastonite for Ca and Si; chromite for Cr; hypersthene for Fe; rhodonite for Mn; enstatite for Mg; albite for Na; and microcline for K. Ca, Si, Mg. Na and K were counted for 20 seconds in the standards, and the rest were counted for more than 50 seconds in order to achieve a total of more than 100,000 counts for each element and thus give reasonable precision. Fifty seconds was the time adopted for sample analysis of all elements except Si and Mg, which were counted for 30 seconds each. The composition of each grain is represented by an average of 3–5 analyses. Little compositional variation (within 1% enstatite content) in individual mineral grains was observed, and no significant zoning was apparent. Thus, no attempt was made to define zonation in the pyroxenes; rather, a defocused beam was used to cover

Table 1. *Chemical composition of clinopyroxene*

Sample	2-E-3	2-E-3	II-89-4	II-89-4	II-89-6	II-89-6	II-16	2-E-6	2-E-6	2-E-6	2-E-7
Grain	n1	n2	n1	n2	n1	n2	n1	n1	n1	n1	n1
	Weight percent oxide										
SiO_2	51.93	51.66	52.59	52.24	52.27	51.98	52.60	51.34	51.31	52.66	51.92
TiO_2	0.76	1.06	0.35	0.40	0.69	0.92	0.55	1.00	1.23	0.61	0.71
Al_2O_3	3.57	3.35	3.32	3.41	3.17	3.54	3.28	3.54	3.59	2.85	3.15
Cr_2O_2	1.08	0.88	1.17	1.15	0.97	1.12	0.96	0.75	0.94	0.63	0.89
FeO	5.46	5.44	5.92	5.53	5.32	5.41	5.81	5.35	5.60	5.70	5.97
MnO	0.07	0.08	0.09	0.08	0.09	0.09	0.10	0.07	0.11	0.11	0.08
CaO	19.25	20.04	17.55	18.63	19.36	19.49	17.39	21.30	19.39	18.66	18.11
MgO	17.52	16.75	18.97	18.30	17.65	17.23	18.98	15.75	17.28	18.48	18.50
K_2O	0.00	0.10	0.00	0.00	0.00	0.01	0.01	0.00	0.01	0.01	0.01
Na_2O	0.39	0.62	0.35	0.32	0.50	0.57	0.40	0.70	0.52	0.55	0.45
Sum	100.04	99.97	100.31	100.07	100.02	100.36	100.07	99.80	99.97	100.27	99.80
	Numbers of cations on the basis of six oxygen atoms										
Si^{4+}	1.896	1.895	1.907	1.903	1.908	1.894	1.910	1.892	1.880	1.915	1.902
Ti^{4+}	0.021	0.029	0.010	0.011	0.019	0.025	0.015	0.028	0.034	0.017	0.019
Al^{3+}	0.154	0.145	0.142	0.147	0.136	0.152	0.140	0.154	0.155	0.122	0.136
Cr^{3+}	0.031	0.025	0.034	0.033	0.028	0.032	0.028	0.022	0.027	0.018	0.026
Fe^{2+}	0.167	0.167	0.180	0.169	0.162	0.165	0.176	0.165	0.172	0.173	0.183
Mn^{2+}	0.002	0.003	0.003	0.003	0.003	0.003	0.003	0.002	0.003	0.003	0.003
Ca^{2+}	0.753	0.787	0.682	0.727	0.757	0.761	0.677	0.841	0.761	0.727	0.711
Mg^{2+}	0.953	0.916	1.026	0.994	0.960	0.936	1.027	0.865	0.944	1.002	1.010
Na^{+}	0.028	0.044	0.025	0.023	0.035	0.041	0.028	0.050	0.037	0.039	0.032
K^{+}	0.000	0.005	0.000	0.000	0.000	0.000	0.000	0.000	0.001	0.000	0.000
Sum	4.005	4.015	4.008	4.008	4.009	4.009	4.005	4.018	4.014	4.017	4.021
	Proportion of end member in the Enstatite-Wollastonite-Ferrosilite Ternary										
En	0.509	0.490	0.543	0.526	0.511	0.503	0.546	0.462	0.503	0.527	0.531
Fs	0.089	0.089	0.095	0.089	0.086	0.089	0.094	0.088	0.091	0.091	0.096
Wo	0.402	0.421	0.361	0.385	0.403	0.409	0.360	0.449	0.406	0.382	0.373
Mg'	0.851	0.846	0.851	0.855	0.855	0.850	0.853	0.840	0.846	0.852	0.847

Mg'=Mg/(Mg+Fe)

Table 1. (*continued*)

Sample	2-E-7	2-E-7	II-W-8	II-W-7	II-5	II-W-20	II-W-20	II-W-20	II-W-20	II-50-1	II-13	II-13
Grain	n2	n3	n1	n1	n1	n1	n2	n3	n4	n1	n1	n2
	Weight percent oxide											
SiO_2	52.37	49.92	51.58	51.40	51.94	52.42	52.25	52.52	50.55	51.11	51.76	51.67
TiO_2	0.62	1.59	0.64	1.03	0.39	0.55	0.37	0.41	1.53	1.26	0.85	0.93
Al_2O_3	3.04	4.43	3.52	3.66	3.17	3.20	3.23	2.44	4.22	3.59	3.38	3.35
Cr_2O_2	0.89	0.83	0.84	1.07	1.09	1.12	1.21	0.67	0.88	0.98	1.06	1.00
FeO	5.92	5.25	6.98	6.41	5.51	5.06	5.41	4.50	5.15	5.82	5.37	5.71
MnO	0.10	0.07	0.15	0.09	0.08	0.08	0.03	0.03	0.06	0.09	0.07	0.08
CaO	17.66	21.24	17.88	18.45	18.72	19.73	18.76	22.07	20.07	19.45	19.57	18.91
MgO	18.83	15.21	17.71	17.54	18.45	17.88	18.47	16.30	16.17	17.08	17.50	17.92
K_2O	0.01	0.01	0.01	0.01	0.01	0.00	0.00	0.00	0.43	0.00	0.00	0.00
Na_2O	0.57	0.84	0.40	0.37	0.29	0.45	0.32	0.69	0.64	0.45	0.42	0.38
Sum	100.01	99.37	99.70	100.04	99.64	100.51	100.05	99.63	99.70	99.85	99.99	99.97
	Numbers of cations on the basis of six oxygen atoms											
Si^{4+}	1.908	1.852	1.896	1.882	1.902	1.904	1.903	1.931	1.863	1.878	1.893	1.890
Ti^{4+}	0.017	0.044	0.018	0.028	0.011	0.015	0.010	0.011	0.042	0.035	0.023	0.026
Al^{3+}	0.131	0.194	0.152	0.158	0.137	0.137	0.139	0.106	0.183	0.156	0.146	0.144
Cr^{3+}	0.026	0.024	0.024	0.031	0.032	0.032	0.035	0.020	0.026	0.029	0.031	0.029
Fe^{2+}	0.180	0.163	0.215	0.196	0.169	0.154	0.165	0.139	0.159	0.179	0.164	0.175
Mn^{2+}	0.003	0.002	0.005	0.003	0.003	0.003	0.001	0.001	0.002	0.003	0.002	0.003
Ca^{2+}	0.689	0.844	0.704	0.724	0.734	0.768	0.732	0.869	0.793	0.766	0.767	0.741
Mg^{2+}	1.023	0.841	0.970	0.957	1.007	0.968	1.003	0.893	0.889	0.936	0.954	0.977
Na^{+}	0.041	0.060	0.028	0.026	0.021	0.032	0.022	0.049	0.046	0.032	0.030	0.027
K^{+}	0.001	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.020	0.000	0.000	0.000
Sum	4.018	4.025	4.013	4.006	4.014	4.013	4.011	4.019	4.023	4.012	4.010	4.011
	Proportion of end member in the Enstatite-Wollastonite-Ferrosilite Ternary											
En	0.540	0.455	0.514	0.510	0.527	0.512	0.528	0.470	0.483	0.498	0.506	0.516
Fs	0.095	0.088	0.114	0.105	0.088	0.081	0.087	0.073	0.086	0.095	0.087	0.092
Wo	0.364	0.457	0.373	0.386	0.384	0.406	0.385	0.457	0.431	0.407	0.407	0.392
Mg'	0.850	0.838	0.819	0.830	0.856	0.863	0.859	0.866	0.848	0.840	0.853	0.848

Mg'=Mg/(Mg+Fe)

Table 2. *Chemical composition of orthopyroxene*

Sample	II-13	II-18	II-50-1	II-5	II-15	II-W-8	II-W-7	II-16	2-E-3	2-E-3	2-E-5	2-E-5
Grain	n1	n1	n1	n1	n1	n1	n1	n1	n1	n2	n1	n2
	Weight percent oxide											
SiO_2	54.99	55.42	54.97	55.31	53.92	53.86	54.50	54.72	55.63	55.07	55.25	55.43
TiO_2	0.36	0.26	0.16	0.29	0.32	0.39	0.28	0.35	0.16	0.25	0.15	0.15
Al_2O_3	1.96	1.55	1.99	1.85	2.71	2.56	2.46	2.30	1.72	2.08	1.92	1.53
Cr_2O_2	0.52	0.55	0.66	0.53	0.64	0.57	0.62	0.61	0.62	0.61	0.63	0.55
FeO	9.53	10.21	9.72	9.52	11.07	11.98	10.87	9.36	9.85	9.58	9.72	10.14
MnO	0.17	0.22	0.14	0.18	0.17	0.21	0.19	0.18	0.16	0.17	0.14	0.19
CaO	2.08	1.98	2.09	2.12	2.26	2.13	2.09	2.13	1.40	1.78	1.68	1.10
MgO	30.46	30.28	30.40	30.60	28.92	28.51	29.43	30.34	30.96	30.62	30.53	30.87
K_2O	0.00	0.00	0.00	0.00	0.00	0.01	0.01	0.01	0.00	0.00	0.00	0.01
Na_2O	0.04	0.05	0.04	0.03	0.04	0.06	0.06	0.05	0.02	0.03	0.07	0.05
Sum	100.11	100.53	100.16	100.43	100.04	100.28	100.51	100.06	100.53	100.18	100.09	100.02
	Number of cations on the basis of six oxigen atoms											
Si^{4+}	1.936	1.948	1.937	1.941	1.914	1.914	1.923	1.928	1.948	1.936	1.944	1.952
Ti^{4+}	0.010	0.007	0.004	0.008	0.008	0.010	0.007	0.009	0.004	0.007	0.004	0.004
Al^{3+}	0.081	0.064	0.082	0.076	0.113	0.107	0.102	0.096	0.071	0.086	0.080	0.064
Cr^{3+}	0.014	0.015	0.018	0.015	0.018	0.016	0.017	0.017	0.017	0.017	0.017	0.015
Fe^{2+}	0.281	0.300	0.286	0.279	0.329	0.356	0.321	0.276	0.289	0.282	0.286	0.299
Mn^{2+}	0.005	0.007	0.004	0.005	0.005	0.006	0.006	0.005	0.005	0.005	0.004	0.006
Ca^{2+}	0.079	0.075	0.079	0.080	0.086	0.081	0.079	0.081	0.052	0.067	0.063	0.041
Mg^{2+}	1.599	1.587	1.596	1.601	1.531	1.510	1.548	1.593	1.616	1.605	1.601	1.621
K^+	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.001	0.000	0.000	0.000	0.000
Na^+	0.003	0.003	0.003	0.002	0.003	0.004	0.004	0.003	0.002	0.002	0.005	0.004
Sum	4.008	4.006	4.010	4.007	4.007	4.006	4.009	4.009	4.004	4.007	4.005	4.006
	Proportion of end member in the Enstatite-Wollastonite-Ferrosilite ternary											
En	0.817	0.809	0.814	0.817	0.787	0.775	0.795	0.817	0.826	0.822	0.821	0.827
Fs	0.143	0.153	0.146	0.143	0.169	0.183	0.165	0.141	0.147	0.144	0.147	0.152
Wo	0.040	0.038	0.040	0.041	0.044	0.042	0.041	0.041	0.027	0.034	0.032	0.021
Mg'	0.851	0.841	0.848	0.851	0.823	0.809	0.828	0.852	0.848	0.851	0.848	0.844

Mg'=Mg/(Mg+Fe)

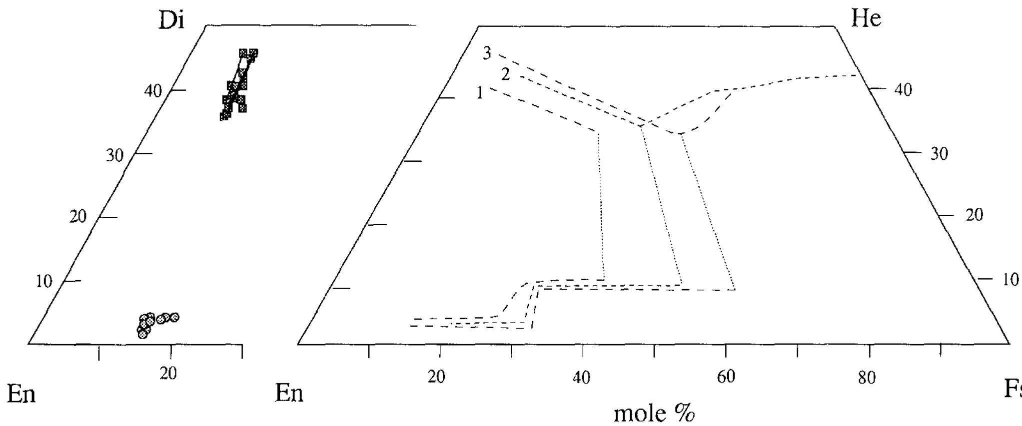

Fig. 1. Quadrilateral plots of pyroxenes. *Left* Pyroxenes from the Jinchuan intrusion. *Right* Pyroxenes from other intrusions for comparison. 1. Jimberlana, 2. Skaergaard, 3. Bushveld. Data other than Jinchuan are from *Campbell* and *Borley* (1974). Dashed lines show the data trends while dotted lines stand for tie line between clino- and ortho-pyroxene

large areas for the analysis and thus generate more representative compositions for the grains. Fine exsolution, even if present, will be averaged in by this procedure.

The calculation of pyroxene stoichiometry based on six oxygens gives the analytical results to within 4 ± 0.025 cations, very close to the predicted value (Table 1, 2). Furthermore, the calculated cation values range from 4.005–4.025 for clinopyroxene and 4.004–4.010 for orthopyroxene. These slightly higher totals for the calculated cations are probably due to the presence of a small amount of Fe^{3+}, which exists in pyroxene crystal structure as a minor component (*Robinson*, 1980). The smaller difference between the calculated and the ideal composition of orthopyroxene compared to that of clinopyroxene suggests that orthopyroxene contains less Fe^{3+} than clinopyroxene, and thus has a lower calculated cation number on a six-oxygen basis.

The plot of pyroxene compositions on the pyroxene quadrilateral (Fig. 1) reveals two distinctive features. First, the Mg′ for both pyroxenes falls within a very narrow ranges for both clinopyroxene (82–87) and orthopyroxene (81–85.5); considerably less compositional variation is present in the Jinchuan than in pyroxenes from well differentiated intrusions such as the Skaergaard and Bushveld (*Atkins*, 1969; *Wager* and *Brown*, 1968), as well as in those from intrusions with less complete differentiation such as the Jimberlana complex (*Campbell* and *Borley*, 1974) and the Great Dyke (*Wilson*, 1982). Second, relatively larger variations of Ca exist in clinopyroxene. This can be seen from the scatter of clinopyroxene compositions along the En–Di join in the composition plot (Fig. 1), which shows that Wo content of the clinopyroxene varies from 36% to 46%. Some samples show two populations of Ca content in pyroxene, a lower Ca clinopyroxene (with 36–40% Wo) and a higher Ca clinopyroxene (with 45–46% Wo). This is illustrated by the connected points in the diagram (Fig. 1).

The concentration of common minor elements (TiO_2, Al_2O_3, Na_2O, Cr_2O_3 and MnO, here they are termed as oxides in order to distinguish them from the calculated cations that are discussed in later sections) in pyroxenes (Table 1, 2) are calculated and plotted in the histograms on Fig. 2. The average values (X) and standard deviation (s) are listed below:

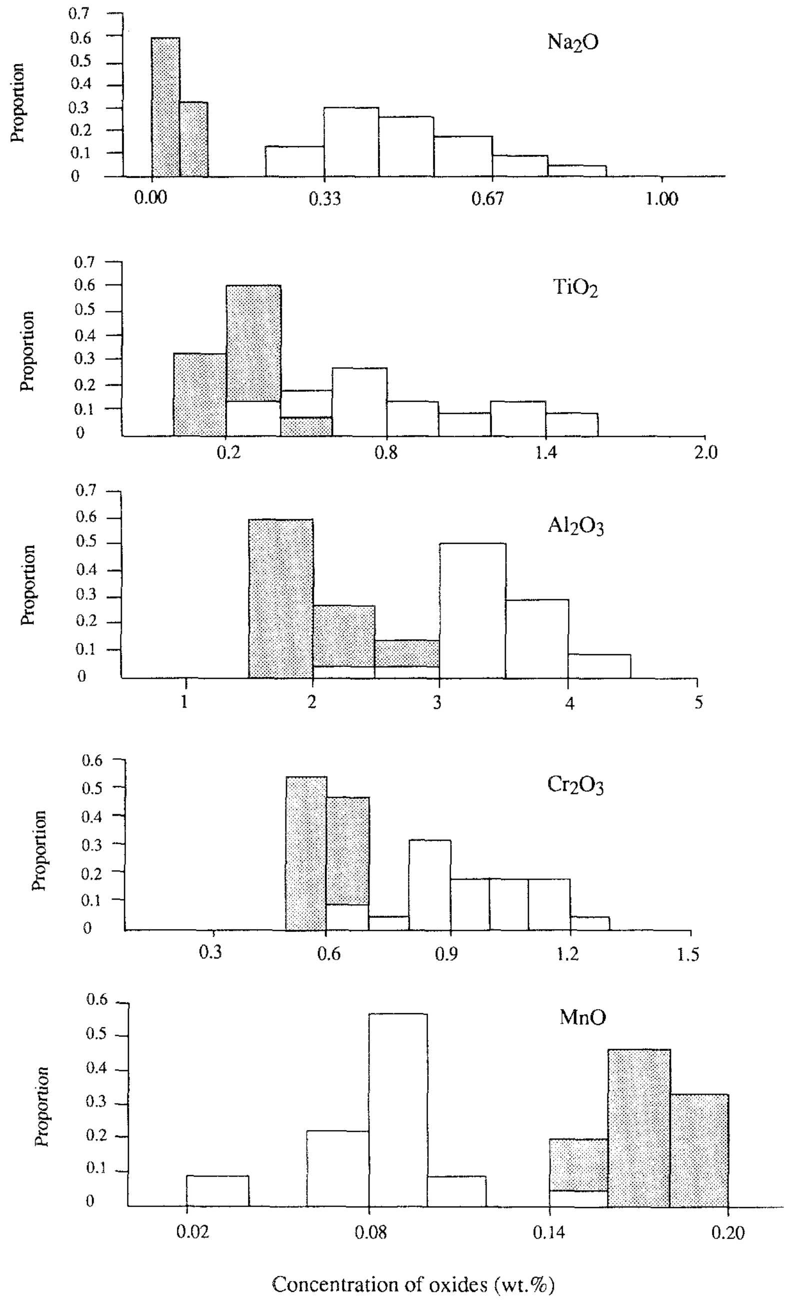

Fig. 2. Histograms showing the distribution of minor elements in pyroxenes. The concentrations are expressed in weight percent. Shaded areas represent minor element distributions of orthopyroxene; open areas represent those of clinopyroxene

	Opx		Cpx	
	X	s	X	s
TiO_2	0.26	0.086	0.80	0.359
Al_2O_3	2.05	0.385	3.39	0.402
Na_2O	0.05	0.012	0.49	0.142
Cr_2O_3	0.59	0.046	0.97	0.156
MnO	0.18	0.025	0.08	0.025

The trivalent and monovalent element oxides such as Al_2O_3, Cr_2O_3 and Na_2O are more highly concentrated in clinopyroxene than in orthopyroxene. The Al_2O_3 content ranges from 2.85 wt.% to 4.43 wt.%, with an average of 3.39 wt.% in clinopyroxene, which is almost twice as much as that in orthopyroxene (2.05 wt.%). The Cr_2O_3 content is from 0.63 to 1.20 wt.% (with an average of 0.97 wt.%) in clinopyroxene in contrast to an average of 0.59 wt.% in orthopyroxene. The content of Na_2O in clinopyroxene averages 0.49 wt.%, which is an order of magnitude higher than in orthopyroxene. The TiO_2 content varies from 0.4 to 1.6 wt.% in clinopyroxene with an average of 0.84, which is about 4 times higher than the level in orthopyroxene. These features reflect the much greater capacity of clinopyroxene to incorporate these minor elements. One exception to this is MnO. The average MnO content in orthopyroxene is 0.18, more than twice the amount in clinopyroxene. This may be due to the fact that Mn^{2+} usually substitutes for Fe^{2+} and should thus be proportional to the FeO content. As FeO is a major component and MnO is a minor component, they are not competing the position in the crystal structure and the higher FeO content means more suitable positions for Mn. Orthopyroxene contains 9.5–12 wt.% FeO, which is about twice the amount in clinopyroxene (5–7 wt.%). This would explain the higher MnO content in orthopyroxene than in clinopyroxene. The higher MnO content of orthopyroxene can also be attributed to the fact that orthopyroxene has twice the number of six co-ordinated positions as clinopyroxene (*Deer* et al., 1978).

Another feature of the minor elements in the Jinchuan pyroxenes is that Al_2O_3 content increases with the decrease of enstatite (En) content, although this variation trend is poorly defined (Fig. 3). The En content generally reflects the nature and differentiation of magma, i.e., a decrease in En should result from an increase of SiO_2 in the magma; Therefore, an increase in Al_2O_3 content in pyroxene corresponds roughly to an increase in SiO_2 of the magma.

Negative correlations between Ti^{4+} and Si^{4+}, and Al^{3+} and Si^{4+} have been recorded in both pyroxenes (Fig. 4). In clinopyroxene, regression of Si^{4+} versus Al^{3+} results in a line with a slope of -1.012 (Fig. 4a), indicating a unit atomic ratio between the increasing amount of Al^{3+} and the decreasing amount of Si^{4+}. Because the discrepancy of tetrahedral Si^{4+} is compensated by Al^{3+} but Al^{3+} occupies both tetrahedral and octahedral sites, the unit linear relationship between Al^{3+} and Si^{4+} requires a relatively constant Al^{3+} in the octahedral site, i.e., the amount of octahedral Al^{3+} does not change with variation of Si^{4+}. In this case, the increased replacement of Si^{4+} by Al^{3+} must be charge balanced by replacement of other high valent cations in the octahedral site. This is well accounted for by the negative linear relationship between Ti^{4+} and Si^{4+} with a slope of around -0.5 (Fig. 4b) and may suggest either that Ti^{4+} is more competitive in occupying octahedral site than Al^{3+},

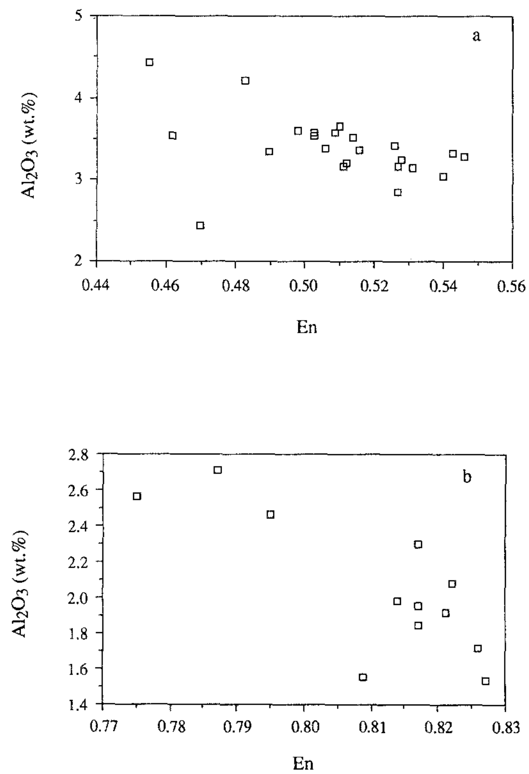

Fig. 3. Variation of Al_2O_3 with the En content of pyroxenes. Both diagrams show negative relationships, indicating a decrease of Al_2O_3 content in pyroxene with the differentiation of the Jinchuan magma. *Upper diagram* clinopyroxene; *lower diagram* orthopyroxene

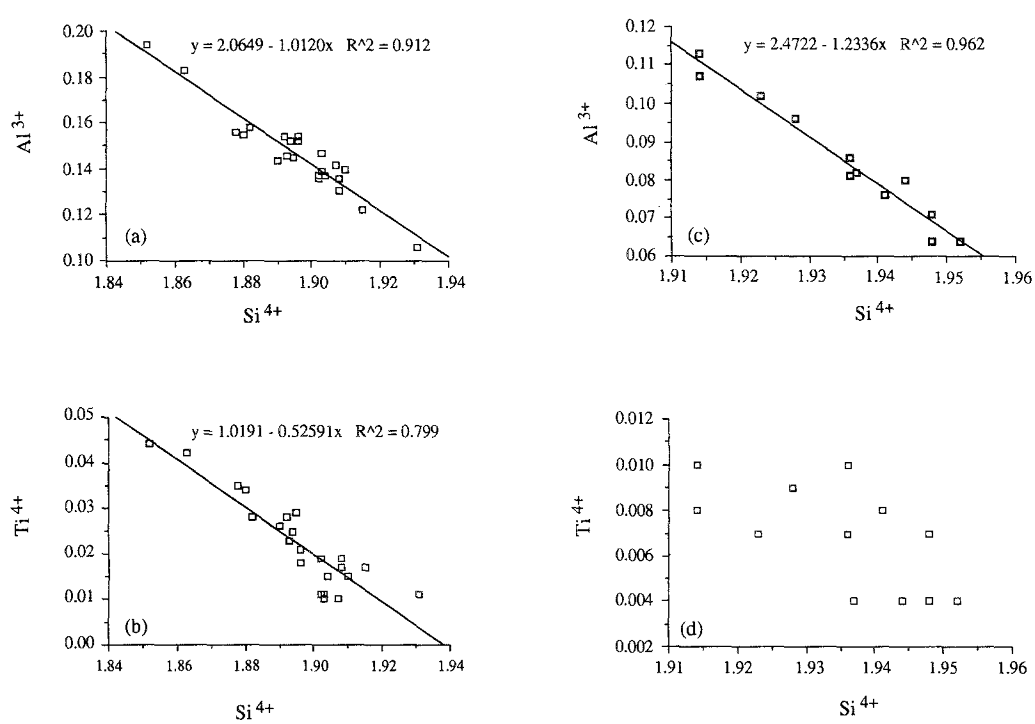

Fig. 4. Plot of the atomic number of Al and Ti versus that of Si in pyroxenes. **a** and **b** represent plots of clinopyroxene; **c** and **d** represent plots of orthpyroxene

or that the replacement of Al^{3+} of Si^{4+} demands an efficient charge balance in the octahedral site. For orthopyroxene, a linear negative relationship between Al^{3+} and Si^{4+} also exists, but a slope (−1.23) of the regression line is greater than that in clinopyroxene (Fig. 4c). This may indicate that both the tetrahedral and octahedral Al^{3+} increase with the decrease of Si^{4+}. On the other hand, the scatter in Ti^{4+} versus Si^{4+} plots (Fig. 4d) may indicate that charge balance is not as critical as structure symmetry.

Factors affecting minor element abundance in pyroxene

Kushiro (1960) and *LeBas* (1962) observed that the Al^{3+} content of natural Ca-rich pyroxenes decreases with increasing SiO_2 content of the magma from which the pyroxenes crystallized. This correlation was also reported in the Bushveld (*Atkins*, 1969) and Jimberlana (*Campbell* and *Borley*, 1974). *Brown* (1967) and *Carmichael* et al. (1970) explained this relationship in terms of the following equation:

$$CaAl_2Si_2O_8 = (CaAl)AlSiO_6 + SiO_2 \tag{1}$$

Thus, the lower the silica activity (a_{SiO_2}) of the magma, the higher the Al^{3+} content of the pyroxenes which crystallized from it. For the Jinchuan pyroxenes, the Al^{3+} content in pyroxene increases with the increase of SiO_2 in the magma, which is the opposite of previous observations (*Campbell* and *Borley*, 1974; *Kushiro*, 1960; *LeBas*, 1962). Also, the Al_2O_3 content of clinopyroxene (2.8–4.4 wt.%) is much higher than that of the Jimberlana clinopyroxene (1.3–2.1 wt.%). This obviously cannot be explained by equation (1) only in terms of variation of SiO_2 in the magma.

Campbell and *Borley* (1974) pointed out that real magmatic systems are much more complicated than that described by equation (1), and that the effect of components such as the oxides of high valent elements (Ti, Al, Cr and Fe) needs to be considered. Thus, the above equation has been revised to the following:

$$0.5Y_2O_3 + 0.5(Al_2O_3) + M_2Si_2O_6 = (MY)AlSiO_6 + SiO_2 + MO \tag{2}$$

or

$$ZO_2 + Al_2O_3 + 2M_2Si_2O_6 = (M_3Z)Al_2Si_2O_{12} + 2SiO_2 + MO \tag{3}$$

where $Y = Ar^{3+}, Cr^{3+}, Fe^{3+}, Z = Ti^{4+}, M = Fe^{2+}, Mg^{2+}Ca^{2+}$. Thus for equation (2):

$$X_{(MY)AlSiO_2} = K_5(X_{Y_2O_3}X_{Al_2O_3})^{0.5}X_{M_2SiO_6}/(X_{SiO_2}X_{MO})$$

Equations (2) and (3) indicate that the substitution of Al^{3+} for Si^{4+} depends not only on SiO_2 activity but also on the activities of other oxides such as TiO_2, Al_2O_3, Cr_2O_3 and Fe_2O_3. Therefore, these factors have been considered to assist in the interpretation of the Jinchuan pyroxenes.

Petrological studies have shown that the Jinchuan intrusive rocks consist of cumulus olivine, and other silicates that are always intercumulus to olivine (*Chai* and *Naldrett*, 1992b). Crystallization of olivine commonly results in a rapid increase of the SiO_2 content of a basaltic magma but a slower increase in SiO_2 would be expected when the magma begins to crystallize pyroxenes. As pyroxenes generally contain greater than 52 wt.% of SiO_2, their crystallization would not change the SiO_2 content of the melt significantly when the magma contained similar amount of SiO_2 when pyroxenes crystallized. Temperature would likely to have changed very little during pyroxene crystallization, thus the a_{SiO_2} would be more or less constant. Even if the activity coefficient of SiO_2 slightly increases with decrease of temperature during pyroxene crystallization (*Campbell* and *Borley*, 1974), the a_{SiO_2} will not increase dramatically. In fact, as will be shown in a later section, the temperature variation was small during the crystallization of Jinchuan pyroxenes, thus the effect of temperature on the a_{SiO_2} was minor. On the other hand, the contents of TiO_2, Al_2O_3, and Cr_2O_3 in olivine and pyroxenes are at minor levels, much lower than in the magma (*Chai* and *Naldrett*, 1992b). Thus, the crystallization of olivine and pyroxenes results in a rapid increase in the TiO_2, Al_2O_3, and Cr_2O_3 contents of the residual magma. The Fe^{3+} content would also have increased rapidly as the above minerals incorporate only, or mostly, Fe^{2+}. However, the oxygen fugacity of the residual melt should be buffered by the presence sulfides in the Jinchuan intrusion. This will cause exchange of Fe^{3+} and Fe^{2+} in the magma. Thus, the Fe^{3+} content probably would not have increased significantly with the crystallization of pyroxenes.

Based on the above discussion, a sharp increase in Al_2O_3 and TiO_2, and more or less constant SiO_2 and Fe_2O_3 would likely occur in the melt during the crystallization of pyroxenes. This would result in an increase in $(MY)AlSiO_2$, i.e., the Al^{3+} content in pyroxene, according to equations (2) and (3).

As for Jimberlana and other layered intrusions such as Skaergaard and Bushveld, this observation is based on Al_2O_3 content of the pyroxenes from the entire layered sequence (*Campbell* and *Borley*, 1974). The greater degree of magma differentiation in these intrusions compared to that of Jinchuan reflects a greater increase in SiO_2 and decrease in temperature. Another major difference between these intrusions and Jinchuan is the presence of plagioclase as a cumulus phase in the former. As can be seen from equation (1), the crystallization of plagioclase will move the reaction toward the left side, and thus reduce the activity of $(CaAl)AlSiO_2$, i.e., the Al_2O_3 content of pyroxenes. According to equation (2), crystallization of plagioclase will reduce the amount of Al^{3+} in the liquid dramatically, which will then lead to a decrease in $(Y_2O_2)^{0.5}(Al_2O_3)^{0.5}/SiO_2$, thus decreasing the amount of $(MY)AlSiO_2$, i.e. Al^{3+}. *Campbell* and *Borley* (1974) have also observed that there is a sharp decrease of Al_2O_3 content in clinopyroxene from the bronzite cumulate to the gabbro layer. The presence of considerablely more plagioclase in gabbro indicates that there must have been a dramatic depletion of Al_2O_3 in the magma, which would have largely reduced the activity of Al_2O_3, thus moving equation (2) to the left side. This can explain the sharp decrease in the amount of Al in pyroxene in gabbro. The more moderate decrease of Al_2O_3 upward in the gabbro may be due to the gradual increase of SiO_2 in the evolved magma (*Campbell* and *Borley*, 1974).

As *Campbell* and *Borley* (1974) stated, the generally lower Al content in the Jimberlana pyroxene (1.3–2.1 wt.%) and higher Al in the Skaergaard pyroxenes (2.8–3.5 wt.%) can be explained by a higher a_{SiO_2} in Jimberlana magma than in that of the Skaergaard. If this is so, the Jinchuan magma may also have had a lower a_{SiO_2} than that of Jimberlana, and even lower than that of the Skaergaard magma, as the Al_2O_3 content of the Jinchuan pyroxenes is higher than in those of the other intrusions. The dominance of clinopyroxene over orthapyroxene in the Jinchuan intercumulus silicate assemblage in contrast to the dominance of bronzite in that of

Jimberlana and the bronzite layer of Skaergaard may reflect a lower SiO_2 activity during the crystallization of the Jinchuan magma, and this seems to support the above interpretation.

Another factor that needs to be considered is temperature. As discussed in a later section, the crystallization temperature of the Jinchuan pyroxenes is generally higher than that of other intrusions. This may have had some effect on the different Al_2O_3 contents of the pyroxenes.

Crystallization temperature of pyroxene

The potential of coexisting high-Ca and low-Ca pyroxenes to yield thermometric information for a wide variety of rocks has long been recognized (*Davis* and *Boyd*, 1966; *Boyd*, 1973; *Wells*, 1977; *Lindsley*, 1983). *Boyd* and *Schairer* (1964) defined a relation between the Ca content of pyroxenes and temperature. *Kretz* (1982) developed a thermometer based on Ca partitioning between augite and low-Ca pyroxene. *Wood* and *Banno* (1973) developed a simple solution model for the Di–En join and presented an analytical expression for temperature based on the estimated enthalpy of the reaction orthoenstatite = clinoenstatite (OEn = CEn). *Wells* (1977) upgraded this thermometer on the basis of reversed experimental data and generated better temperature information, particularly for the high temperature pyroxenes. Both of these thermometers have been widely used (*Wilson*, 1982; *Gole* et al., 1990), although their assumptions of ideal mixing in clinopyroxene and a large reaction enthalpy for OEn = CEn are generally regarded as erroneous (*Holland* et al., 1979; *Lindsley* et al., 1981; *Davison* et al., 1982). At high temperatures, the errors introduced by these assumptions tend to be self-cancelling, but the calculated temperatures are 100 °C or more too high for metamorphic pyroxenes (*Lindsley*, 1983). Therefore, they may be useful only for igneous intrusions.

Lindsley (1983) combined experimental Ca–Mg–Fe pyroxene phase relations determined at 800–1200 °C and from less than one atmosphere up to 15 kb pressure to produce a graphical two-pyroxene thermometer which he claimed to be applicable to wide variety of rocks with the limitation that pyroxenes must have $Wo + En + Fs > 90$

The temperatures of the Jinchuan pyroxene pairs were calculated using the methods of *Lindsley* (1983), *Wood* and *Banno* (1973) and *Wells* (1977). The *Lindsley* (1983) method shows two sets of temperatures; one group related to a lower Ca content in clinopyroxene gave a temperature range of about 1180–900 °C; the other related to a higher Ca content in clinopyroxene gave a temperature range around 750–600 °C. The lower temperature pyroxene may reflect a secondary phase because it is always associated with a high temperature primary pyroxene. The temperatures calculation according to the *Wood* and *Banno* (1973) and *Wells* (1977) methods yielded similar values with a difference of less than 40 °C (Table 3). The *Wells* (1977) method gave a slightly larger temperature span (1069–1231 °C) than the *Wood* and *Banno* (1973) method (1108–1229 °C). Generally, the temperatures calculated by the *Wood* and *Banno* (1973) and *Wells* (1977) method are about 100 °C higher than those determined by *Lindsley*'s (1983) method, and the temperature range of the former is much narrower than that of the latter.

Lindsley (1983) estimated the temperature variation using his graphic geothermometer for the MgO-rich Skaergaard augite. When the original composition was

Table 3. *Crystallization temperature of pyroxenes*

Sample	$X_{Fe_{opx}}$	$a_{Mg_{cpx}}$	$a_{Mg_{opx}}$	ln (a_{cpx}/a_{opx})	T * (°C)	T ** (°C)	ABS (T *-T **)
2-E-3	0.149	0.142	0.650	-1.521	1148.9	1128.0	20.9
2-E-3	0.145	0.110	0.639	-1.759	1108.1	1069.5	39.6
II-16	0.143	0.194	0.629	-1.176	1229.3	1231.2	1.9
II-w-8	0.180	0.163	0.567	-1.247	1164.3	1183.3	19.0
II-w-7	0.165	0.155	0.594	-1.343	1163.9	1166.1	2.2
II-5	0.144	0.166	0.636	-1.343	1191.9	1180.8	11.1
II-50-1	0.145	0.129	0.630	-1.586	1141.3	1113.5	27.8
II-13	0.145	0.135	0.634	-1.547	1149.0	1123.8	25.2
II-13	0.145	0.153	0.634	-1.422	1174.2	1157.9	16.3

T * : Temperature calculated by Wood and Banno (1973)'s method and T ** by Wells (1977)'s method

used, it showed a temperature range from 1050–1100 °C for the most MgO-rich and FeO-rich augite, respectively. Thus, temperature apparently increased with crystallization. *Lindsley* and *Andersen* (1983) presented extensive textural, chemical and experimental evidence to demonstrate that the Mg-rich augites from the Skaergaard have undergone a significant amount of granule exsolution, as have the pigeonites. Therefore, special treatment was suggested to be used to process the original composition (*Lindsley*, 1983), which produced temperatures for the MgO-rich pyroxenes slightly higher than those based on the original composition. The temperatures estimated by *Lindsley* (1983)'s method were based on projection of the original pyroxene composition of Jinchuan intrusion. These may not represent the actual temperatures at which they crystallized. On the other hand, the *Wood* and *Banno* (1973) and *Wells* (1977) methods gave very similar temperatures, and these temperatures are thus adopted to represent the crystallization temperatures of the pyroxenes.

Mg^{2+}–Fe^{2+} distribution between clinopyroxene and orthopyroxene

The thermodynamic basis for the distribution of Mg^{2+} and Fe^{2+} in coexisting orthopyroxene and clinopyroxene was first considered by *Kretz* (1961) and *Bartholomé* (1962), following the discovery of *Mueller* (1960, 1961) that for a number of coexisting pyroxene pairs, a plot of $X_{Mg}/X_{Fe} = Mg^{2+}/(Mg^{2+} + Fe^{2+})$ in orthopyroxene against that in clinopyroxene gives a curve of the type to be expected if both minerals are an ideal or near ideal mixture.

Kretz (1963) presented data pertaining to the distribution coefficients (Kd) for coexisting ortho- and clinopyroxene from twenty-five high grade metamorphic and fifteen igneous rocks, and showed that Kd varies between 0.51 and 0.65 for the pyroxene pairs of the first group and from 0.65 to 0.86 for the second group. *Saxena* (1968) also studied the distribution of Mg^{2+} and Fe^{2+} between coexisting orthopyroxene and Ca-pyroxenes in igneous and metamorphic rocks and presented Kd values of 0.54 and 0.73 for the metamorphic and igneous rocks, respectively.

However, distribution coefficients are dependent on T, P and, perhaps in some cases, on the concentrations of other elements (*Atkins*, 1969), and should be used only to indicate physico-chemical conditions rather than to discriminate between igneous and metamorphic assemblages. For example, in previously investigated igneous assemblages of pyroxenes, emphasis was mainly on layered intrusions in which the pyroxenes were characterized by normal Fe-enrichment differentiation trends. Other factors, such as minor element contents and oxygen fugacity, may have been overlooked.

The Jinchuan pyroxenes are different from those of the other intrusions in the following aspects:

1) All pyroxenes are interstitial to cumulus olivine.

2) Small Fe-enrichment is seen in the two pyroxene field of the Ca–Mg–Fe diagram (Fig. 1); on the contrary, Ca-enrichment is seen in the later formed Ca-pyroxene.

3) The pyroxenes are generally associated with sulfide mineralization, which may reflect more reducing conditions.

4) Al_2O_3 content in the Jinchuan pyroxenes is generally higher than that of the other intrusions and may have affected Kd values.

Therefore, the Kd values of the Jinchuan pyroxenes are of particular importance as they can provide data on pyroxenes under different geological environments.

Definitions of Kd vary in different publications (*Brown*, 1967; *Atkins*, 1969; *Kretz*, 1982). In this work, Kd is defined as follows:

$$Kd = (X_{Fe^{2+}}/X_{Mg^{2+}})_{cpx}/(X_{Fe^{2+}}/X_{Mg^{2+}})_{opx} \quad (4)$$

Pyroxenes may contain certain amounts of Fe^{3+}, and the estimation of Fe^{3+} from the electron microprobe analyses depends on the method of calculation and the quality of the analyses (*Finger*, 1972; *Ryburn* et al., 1976). In this work, we have calculated the Fe^{3+} content using the charge balance method (*Ghent* and *Coleman*, 1973; *Ryburn* et al., 1976). This has resulted in an average Fe^{3+}/TFe of 0.13 for clinopyroxenes and 0.05 for orthopyroxenes. The Fe^{2+} content of pyroxenes is then calculated by subtracting the amount of Fe^{3+}, using the average Fe^{3+}/TFe value, from the total iron content by the charge balance method. The reason for using the average ratios is that they reduce the uncertainty of the Fe^{3+} calculations of individual analyses and that pyroxenes crystallized under similar conditions are likely to have similar proportions of Fe^{3+}.

A linear relationship is evident for X_{FeO}/X_{MgO} in pyroxene pairs from the Jinchuan intrusion (Fig. 5). This gives a distribution coefficient of Mg^{2+} and Fe^{2+} between clinopyroxene and orthopyroxene of 0.86. The linear relationship also implies that alteration and/or metamophism have not affected the chemical composition of the fresh pyroxenes, or at least not substantially.

This calculated value is greater than those obtained for other igneous rocks such as the Bushveld complex (*Atkins*, 1969), the Skaergaard intrusion (*Wager* and *Brown*, 1968) and the Great Dyke (*Wilson*, 1982). This value may reflect either the higher temperature nature of the Jinchuan pyroxenes, or significantly different conditions in the Jinchuan magma, such as pressure, composition and oxygen or sulfur fugacities.

Although *Atkins* (1969) emphasized the importance of pressure on Kd values for the Bushveld complex, others have tended to neglect its effect based on wide ranges of experimental and natural observations (*Davis* and *Boyd*, 1966; *Wood* and *Banno*,

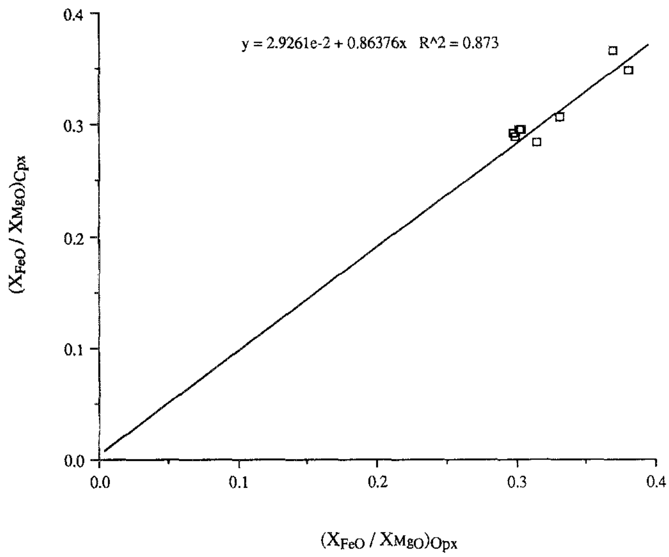

Fig. 5. Plot of the correlation of FeO/MgO ratio of the pyroxenes. The slope of the regression line may represent a Kd of 0.86 between clinopyroxene and orthopyroxene. *Cpx* clinopyroxene, *Opx* orthopyroxene

1973; *Kretz*, 1982). It is difficult to determine the pressures under which pyroxenes crystallized, but it seems unlikely that great differences in pressure existed between the Jinchuan and other intrusions. Thus, the influence of the pressure can be excluded.

Temperature is generally accepted as an important factor in controlling the Kd of pyroxene pairs (*Naldrett* and *Kullerud*, 1967; *Wood* and *Banno*, 1973; *Wells*, 1977; *Kretz*, 1982). The calculated temperature of the Jinchuan pyroxenes is higher than those of the other intrusions, reflecting early stage crystallization from a basaltic magma (Table 3). Thus, the effect of temperature will be considered first. If this cannot account for the variation of Kd, the effect of minor elements such as Al, Ti and Cr and possibly other factors (f_{O_2}) may then be considered, as both these minor elements and the major elements (Mg and Fe) occupy the same octahedral sites in pyroxenes.

The exchange reaction of Mg^{2+}–Fe^{2+} between associated ortho- and clinopyroxene crystals can be shown as follows:

$$FeSiO_3 + CaMgSi_2O_6 = MgSiO_3 + CaFeSi_2O_6 \tag{5}$$

At equilibrium,

$$K = (a_{CaFeSi_2O_6} a_{MgSiO_3})/(a_{CaMgSi_2O_6} a_{FeSiO_3}) \tag{6}$$

The activity terms are difficult to evaluate in the natural system (*Wells*, 1977). The two pyroxenes may behave as near ideal two-site solutions of $CaMgSi_2O_6$ and $MgSiO_3$ in the case of igneous system (*Wood* and *Banno*, 1973; *Wells*, 1977) although this is not generally accepted (*Saxena*, 1976; *Holland* et al., 1979). If so, the equilibrium coefficient may be expressed as:

$$\begin{aligned} K &= (a_{CaFeSi_2O_6} \cdot a_{MgSiO_3})/(a_{CaMgSi_2O_6} \cdot a_{FeSiO_3}) \\ &= (\gamma_{CaFeSi_2O_6} \gamma_{MgSiO_3})/(\gamma_{CaMgSi_2O_6} \gamma_{FeSiO_3}) \\ &\quad (X_{CaFeSi_2O_6} X_{MgSiO_3})/(X_{CaMgSi_2O_6} X_{FeSiO_3}) \\ &= (X_{CaFeSi_2O_6} X_{MgSiO_3})/(X_{CaMgSi_2O_6} X_{FeSiO_3}) = Kd \end{aligned} \tag{7}$$

The equilibrium conditions for the above relation can be expressed as follows.

$$\Delta G_{p,t} = -RT \ln K$$
$$\ln K = \ln Kd = -\Delta G/RT$$
$$= -\Delta H/RT + \Delta S \quad (8)$$

The latter equation defines a linear relationship between ln Kd and 1/T if ΔH is independent of temperature. *Wood* and *Banno* (1973) indicated that for the reaction of Mg-orthopyroxene and clinopyroxene, ΔH and ΔS can be considered to be independent of temperature. Similarly, for the Fe-orthopyroxene and clinopyroxene reaction, it is reasonable to assume that the ΔH and ΔS are independent of temperature. Therefore, the ΔH and ΔS of the above reaction can also be considered to be independent of temperature.

Kd data calculated on pyroxene pairs from different occurrences, together with the calculated temperature based on the thermometry of *Wells* (1977) or the experimental data, are listed below.

	Kd	T (°C)	Ref.
Rhyolite	0.557	965	*Carmichael* (1967b)
Bushveld	0.669	1029	*Arkins* (1969)
Skaergaard	0.736	1100	*Kretz* (1982)
The Great Dyke	0.75	1069	*Wilson* (1982)
Jinchuan	0.86	1150	This work
Lherzolite Nodule	0.98	1347	*Nixon* and *Boyd* (1974)
Experiment	1.05	1334	*Green* and *Ringwood* (1967)

The analyses for the latter two pyroxene pairs were made by microprobe analyses with no attempt to distinguish between Fe^{3+} and Fe^{2+} (*Nixon* and *Boyd*, 1974; *Green* and *Ringwood*, 1967). The Fe^{2+} contents were artificially adjusted to 85% and 95% of the total iron for clinopyroxene and orthopyroxene, respectively, for consistency.

The calculated data are plotted on a Kd versus 1000/T diagram (Fig. 6). An excellent linear trend can immediately be seen with a R factor of greater than 0.99. A least squares fit to their points yields the following relationship between Kd and 1/T:

$$\ln Kd = -2109.2/T + 1.6463 \quad (9)$$

The above calculation and discussion have clearly assigned two significant meanings to the Kd values: 1) Kd is controlled primarily, if not solely, by temperature; 2) The equation may be used as a geothermometer.

The Kd value of the Jinchuan pyroxenes falls almost precisely on the regression line defined by the pyroxene data from other intrusions and experiments, and is exactly what is predicted by an ideal mixing model. Therefore, the Kd of the Jinchuan pyroxene must have been controlled entirely by temperature. *Wells* (1977) plotted data on orthopyroxene and clinopyroxene with large variations of Al_2O_3 content (up to 12 wt.%), and found that ln Kd is not related consistently to the amount of Al_2O_3. In addition, although the crystallization conditions for the

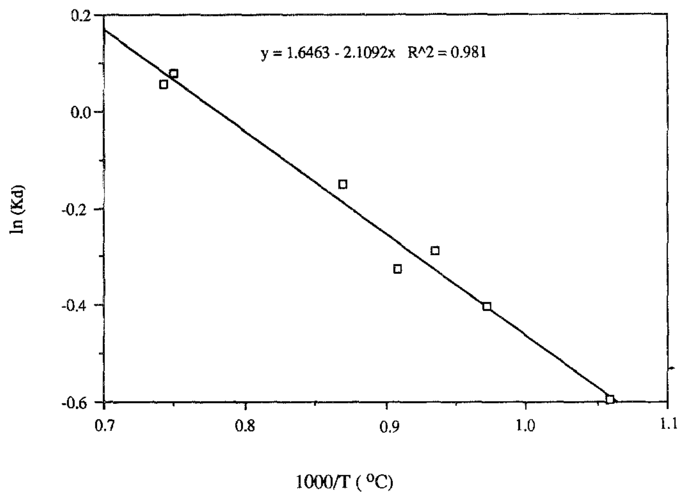

Fig. 6. Plot of Ln (Kd) versus 1000/T (°C) of pyroxenes from selected intrusions. A linear relationship is observed, indicating temperature is the predominant controlling factor on the distribution coefficient of pyroxenes

Jinchuan pyroxenes are significantly different from those of the other intrusions, the Kd value still falls on the same linear trend. This implies that these conditions, such as compositions, oxygen and sulfur fugacities of the magma, have little effect on the distribution coefficient.

Kretz (1982) developed a Kd thermometer based on some experimental data, but actual pyroxene compositions do not fall onto the line defined by his formula. Equation (9) indicates that Kd is linearly correlated with crystallization temperatures for pyroxenes from various intrusions. This suggests that some parameters in *Kretz*'s (1982) equation may not have been correctly defined. However, it is beyond the scope of this study to establish a geothermometer; this would require further experimental investigation and theoretical modeling.

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